Bartomeu Garau Verger

Research Support Technician

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Education

2023-2024 | **Master's Degree in Advanced Mathematics and Mathematical Engineering**
Universitat Politècnica de Catalunya • Barcelona, Spain
Focusing on the *Differential Equations* and *Modeling in Engineering and Biomedical Sciences* fields.
Average : 9.21/10
Master's thesis : *Calderón-Zygmund estimates for the Poisson equation* (Mark : 9.5/10). Supervised by Clata Torres-Latorre and Xavier Ros-Oton.

2019-2023 | **Bachelor's Degree in Mathematics**
Universitat de les Illes Balears • Palma, Spain
Average : 7.59/10 (Second ranked)
Bachelor's thesis : *Variational models for underwater image enhancement* (Mark : 10/10). Supervised by Catalina Sbert.

2017 - 2019 | **Scientific Baccalaurate**
Universitat Politècnica de Catalunya • Barcelona, Spain
Col·legi Sant Antoni Abat • Palma, Spain
Average : 10/10 (First ranked)
University Entrance Exam (PBAU) : 9.96/10, 13.86/14 (Second ranked)

Research experience

Present October 2024 | **Research Support Technician**
Intitute of Applied Computing & Community Code • Palma, Spain
- Maintenance of the Institute's web presence and promotion
- Organisation of scientific outreach conferences and seminars
- Collaboration in different research activities

July 2022 September 2021 | **Research intern**
Universitat de les Illes Balears • Palma, Spain
- Undergraduate research intern
- Development of algorithm for the minimization of energy functionals
- Implementation of deep unfolding networks
- Applications to image dehazing and denoising

Supervised by Prof. Bartomeu Coll, Joan Duran, Catalina Sbert and Julia Navarro

Articles

Introducing nonlocal elliptic equations and nonlocal physics-informed networks for image restoration 2025
Bartomeu Garau, Catalina Sbert and Joan Duran. *2025 IEEE International Conference on Image Processing (ICIP)*
Status : Submitted.
Nonlocal PDE | CNN | PINN | denoising

Two Simple Unfolded Residual Networks for Single Image Dehazing 2025
Bartomeu Garau, Joan Duran and Catalina Sbert. *International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (VISIGRAPP), 2025*
Status : Accepted
Image Dehazing | Deep Learning | Unfolding | Residual Network | Channel Attention | Variational Methods

Conferences and seminars

20th International Conference on Computer Vision Theory and Applications (VISAPP 2025) — February 2025
Oporto, Portugal
Speaker on the VISAPP 2025 presenting the short paper *Two Simple Unfolded Residual Networks for Single Image Dehazing*. More information on the following link.

Harmonic Measure and Free Boundary Problems — Fall 2023
BGSMath Graduate School • Barcelona, Spain
Assistance to the course on Harmonic Measure and Free Boundary Problems. Duration : 20h. More information on the following link.

Barcelona Analysis Conference (BAC24) — June 2024
Barcelona Analysis Seminar • Barcelona, Spain
Assistance to the BAC24, a Satellite Activity of the ECM 2024. More information on the following link.

English in Margate - Standard Course — Summer 2018
Margate, Kent, England
Assistance to the English in Margate Course. Duration : 30h.

Invited talks

23/10/2024 *Regularity Theory for Linear and Nonlinear Poisson-Type Equations*
Institute of Applied Computing & Community Code • Palma, Spain

Outreach activities

Talks : "Els matemàtics tornen als seus instituts"
Sessions where the students of the B.Sc in Mathematics explain to high school students what higher level mathematics are and what we do in our Bachelor. Sessions given on 2023 and 2024 at Col·legi Sant Antoni Abat, Palma

Talks : "T'ajudam a decidir"
Online sessions to inform high schools students about how the B.Sc. in Mathematics works and what we do there. Sessions given for the academic years 2021-22, 2022-23, 2023-24, 2024-25

Podcast : "El racó de Hilbert" — 2022-2023
Co-founder, member and editor of *El racó de Hilbert* (during its first season), a podcast made by students from the B.Sc. in Mathematics at the University of the Baleric Islands, where we talk about mathematics, science, and a great variety of topics. Available at Spotify

Scientific groups

Present – December 2024 | **Mathematics, Imaging and Learning**
Member of the Mathematics, Imaging and Learning group from the Universitat de les Illes Balears.

Present – October 2024 | **Institute of Appliead Computing & Community Code**
Member of the Institute of Appliead Computing & Community Code

Positions of responsibility

2023 – 2020 | **Student's Representative**
Universitat de les Illes Balears • Palma, Spain
Member of the Quality Assurance Comission of the Bachelor's Degre in Mathematics as the students' representative.

2023 – 2019 | **Class Representative**
Universitat de les Illes Balears • Palma, Spain

Programming tools

Python	Advanced (with emphasis on computer vision and deep learning tools)
Matlab	Intermediete (with emphasis on numerical methods for partial differential equations)
R	Intermediate
Git	Basic
LaTeX	Advanced

Langues

- **Catalan :** Native, C1
- **Spanish :** Native
- **English :** Professional knowledge, B2

Soft skills

Leadership • Creativity • Eye for details • Autonomy • Patience • Ablitiy to plan and organize